

HITOYOSHI, Japan

WMO: 478240

Lat: 32.2181N

Lon: 130.7553E

Elev: 148

StdP: 99.56

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) -2.2	(d) -8.1	(e) 1.9	(f) 4.4	(g) -6.6	(h) 2.2	(i) 3.7	(j) 6.3	(k) 5.7	(l) 5.5	(m) 6.0	(n) 0.9	(o) 60	(p) 0.339

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.7	(c) 33.6	(d) 25.2	(e) 32.6	(f) 24.9	(g) 31.5	(h) 24.4	(i) 26.3	(j) 30.9	(k) 25.7	(l) 30.1	(m) 25.3	(n) 29.4	(o) 2.6	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	27.9	24.7	20.0	27.5	24.2	19.5	27.1	82.8	31.0	80.5	30.2	78.6	29.5	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.9	(b) 4.9	(c) 4.1	(e) -5.6	(f) 35.1	(g) 1.4	(h) 0.8	(i) -6.7	(j) 35.7	(k) -7.5	(l) 36.2	(m) -8.3	(n) 36.7	(o) -9.3	(p) 37.3		
(a) 5.9	(b) 4.9	(c) 4.1	(e) -6.0	(f) 27.6	(g) 1.4	(h) 1.1	(i) -7.1	(j) 28.4	(k) -7.9	(l) 29.0	(m) -8.7	(n) 29.6	(o) -9.7	(p) 30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	5.1	6.7	10.1	15.0	19.5	22.9	26.8	27.4	24.4	18.9	12.6	7.0
	DBStd	8.23	2.72	3.56	3.33	3.19	2.24	2.22	1.74	1.44	2.46	3.06	3.66	3.25
	HDD10.0	424	154	104	41	3	0	0	0	0	0	0	15	106
	HDD18.3	1669	411	324	257	106	13	1	0	0	0	29	174	353
	CDD10.0	2764	2	13	43	153	296	386	520	540	432	276	93	12
	CDD18.3	967	0	0	0	6	51	136	262	282	183	47	2	0
	CDH23.3	7688	0	0	2	56	356	799	2318	2629	1289	234	5	0
	CDH26.7	2653	0	0	0	4	59	194	895	1053	416	31	0	0
Wind	WSAvg	1.4	1.4	1.4	1.5	1.5	1.5	1.4	1.4	1.5	1.3	1.1	1.1	1.2
Precipitation	PrecAvg	2492	74	106	160	182	210	550	417	247	250	116	99	82
	PrecMax	4099	182	203	404	365	371	959	1053	766	530	353	212	190
	PrecMin	1493	4	47	49	53	63	178	68	35	26	15	27	7
	PrecStd	502	36	49	72	77	87	205	240	146	144	77	48	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	20.0	23.0	27.2	30.2	31.5	34.7	34.9	33.1	29.4	24.0	18.0
		MCWB	11.0	14.7	14.5	17.1	18.9	22.8	25.6	25.9	24.7	22.4	18.3	13.8
	2%	DB	13.3	17.1	20.4	25.0	28.2	30.1	33.6	33.8	31.7	27.4	21.7	15.8
		MCWB	9.5	12.8	13.1	16.1	18.2	22.3	25.2	25.3	23.8	20.7	16.9	12.8
	5%	DB	11.8	15.0	18.2	23.2	26.9	28.9	32.6	32.9	30.6	25.9	20.1	14.1
		MCWB	8.4	11.1	12.3	15.3	17.9	21.8	24.9	25.0	23.4	19.7	16.1	11.2
	10%	DB	10.3	13.0	16.3	21.4	25.4	27.6	31.4	31.8	29.3	24.4	18.5	12.5
		MCWB	7.2	9.3	11.6	15.1	17.6	21.5	24.7	24.8	23.0	19.1	15.5	9.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	16.1	16.9	20.7	23.0	25.2	27.1	27.2	26.1	24.1	20.2	15.5
		MCDB	14.3	18.0	18.7	22.6	25.7	28.4	32.6	32.4	30.4	27.1	22.1	16.6
	2%	WB	11.0	14.0	15.4	19.2	21.8	24.6	26.2	26.5	25.3	22.8	18.5	13.5
		MCDB	12.0	16.2	17.8	21.3	24.0	27.7	30.9	31.3	29.0	25.4	20.0	14.8
	5%	WB	9.1	12.0	14.1	17.6	20.7	23.9	25.7	25.9	24.7	21.7	17.3	12.0
		MCDB	10.9	14.0	16.8	20.5	23.1	26.8	30.0	30.4	28.1	24.2	19.1	13.6
	10%	WB	7.7	10.1	12.7	16.4	19.6	23.3	25.2	25.5	24.1	20.4	16.0	10.4
		MCDB	9.6	12.1	15.3	19.4	22.6	25.6	29.3	29.8	27.3	22.8	17.8	12.0
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.3	10.9	11.5	11.1	8.0	8.3	8.7	9.2	10.3	9.9	9.4
		MCDBR	11.7	12.8	14.2	14.9	14.3	10.9	10.7	10.6	10.6	11.7	11.1	10.7
		MCWBR	7.7	8.0	7.9	7.0	5.9	4.1	3.7	3.6	3.9	5.4	6.2	7.0
	5% WB	MCDBR	9.2	9.7	10.1	9.4	8.4	6.9	9.0	8.9	8.3	8.1	7.6	8.4
		MCWBR	7.2	7.2	7.1	5.7	5.0	3.4	3.6	3.4	3.5	4.2	5.0	6.5
Clear-Sky Solar Irradiance	taub	0.413	0.451	0.509	0.529	0.539	0.543	0.536	0.520	0.497	0.453	0.415	0.403	
	taud	2.117	2.047	1.897	1.903	1.945	2.019	2.078	2.115	2.141	2.182	2.207	2.163	
	Ebn at Noon	779	789	772	777	771	765	768	775	776	780	774	766	
	Edh at Noon	128	151	189	195	189	175	164	156	146	131	116	116	
All-Sky Solar Radiation	RadAvg	2.47	3.10	4.09	4.79	5.03	4.08	5.05	5.17	4.32	3.64	2.75	2.29	
	RadStd	0.24	0.27	0.27	0.46	0.56	0.66	0.63	0.43	0.47	0.40	0.28	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page